Math 431 – Spring 2026

Homework 10 – Solutions

7.2 The possible values for both X and Y are 0 and 1, hence $X+Y$ can take the values $0,1$ and 2. If $X+Y=0$ then we must have $X=0$ and $Y=0$ and by independence we get

$$P(X+Y=0)=P(X=0,Y=0)=P(X=0)P(Y=0)=(1-p)(1-r).$$

Similarly, if $X+Y=2$ then we must have $X=1$ and $Y=1$:

$$P(X+Y=2)=P(X=1,Y=1)=P(X+1)P(Y=1)=pr.$$

We can now compute $P(X+Y)=1$ by considering the complement:

$$P(X+Y=1)=1-P(X+Y=0)-P(X+Y=2)=1-(1-p)(1-r)-pr=p+r-2pr.$$

We could have also used the fact that $X+Y=1$ can only happen if $X=0, Y=1$ or $X=1, Y=0$:

$$P(X+Y=1)=P(X=0,Y=1)+P(X=1,Y=0)=(1-p)r+p(1-r)=p+r-2pr.$$

We have computed the probability mass function of $X+Y$ which identifies its distribution.

7.16 The probability mass function of X is

$$p_X(k)=\frac{\lambda^k}{k!}e^{-\lambda}, \qquad k=0,1,2,\dots$$

while the probability mass function of Y is $p_Y(0)=1-p$, $p_Y(1)=p$. Using the convolution formula we get

$$p_{X+Y}(n)=\sum_k p_X(k)p_Y(n-k).$$

The possible values of $X+Y$ are $0,1,2,\dots$, so we only need to deal with $n\geq 0$. We only have $p_Y(n-k)\neq 0$ if $n-k=0$ or $n-k=1$ so we get

$$p_{X+Y}(n)=p_X(n)p_Y(0)+p_X(n-1)p_Y(1).$$

If $n=0$ then $p_X(n-1)=0$, so

$$p_{X+Y}(0)=p_X(0)p_Y(0)=(1-p)e^{-\lambda}.$$

For $n>0$ we get

$$\begin{aligned} p_{X+Y}(n)&=p_X(n)p_Y(0)+p_X(n-1)p_Y(1)=(1-p)\frac{\lambda^n}{n!}e^{-\lambda}+p\frac{\lambda^{n-1}}{(n-1)!}e^{-\lambda}\\ &=\lambda^{n-1}\frac{(\lambda(1-p)+np)}{n!}e^{-\lambda}. \end{aligned}$$

Thus the probability mass function of $X+Y$ is

$$p_{X+Y}(n)=\begin{cases}(1-p)e^{-\lambda}, & \text{if } n=0,\\ \lambda^{n-1}\frac{(\lambda(1-p)+np)}{n!}e^{-\lambda}, & \text{if } n\geq 1.\end{cases}$$

7.20 (b) Note that X takes values in $(0,1)$, Y takes values in $(1,2)$ so $X+Y$ will take values in $(1,3)$. For a given $z \in (1,3)$ the convolution formula gives

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx = \int_0^1 f_X(x) f_Y(z-x) dx,$$

where we used the fact that $f_X(x) = 0$ outside $(0,1)$. For a given $1 < z < 3$ the function $f_Y(z-x)$ is nonzero if and only if $1 < z - x < 2$, which is equivalent to $z - 2 < x < z - 1$. Since we must have $0 < x < 1$ for $f_X(x)$ to be nonzero, this means that $f_X(x) f_Y(z-x)$ is nonzero only if $\max(0, z-2) < x < \min(1, z-1)$. Thus

$$\begin{aligned} f_{X+Y}(z) &= \int_0^1 f_X(x) f_Y(z-x) dx = \int_{\max(0,z-2)}^{\min(1,z-1)} 2x dx = x^2 \Big]_{\max(0,z-2)}^{\min(1,z-1)} \\ &= \min(1, z-1)^2 - \max(0, z-2)^2. \end{aligned}$$

We can evaluate the min and the max explicitly by separating the $1 < z \le 2$ and $2 < z < 3$ cases. If $1 < z \le 2$ then

$$\min(1, z-1) = z - 1, \qquad \max(0, z-2) = 0.$$

If $2 < z < 3$ then

$$\min(1, z-1) = 1, \qquad \max(0, z-2) = z - 2.$$

Hence

$$f_{X+Y}(z) = \begin{cases} (z-1)^2, & \text{if } 1 < z \le 2, \\ 1 - (z-2)^2, & \text{if } 2 < z < 3, \\ 0, & \text{otherwise.} \end{cases}$$

7.24 Suppose that the variances of X, Y and Z are σ_X^2, σ_Y^2 and σ_Z^2. Using Fact 7.9 about the linear combination of independent normal random variables we have that

$$X + 2Y - 3Z \sim \mathcal{N}(0, \sigma_X^2 + 4\sigma_Y^2 + 9\sigma_Z^2),$$

and $\frac{X+2Y-3Z}{\sqrt{\sigma_X^2+4\sigma_Y^2+9\sigma_Z^2}} \sim \mathcal{N}(0,1)$.

This gives

$$P(X + 2Y - 3Z > 0) = P\left(\frac{X + 2Y - 3Z}{\sqrt{\sigma_X^2 + 4\sigma_Y^2 + 9\sigma_Z^2}} > 0\right) = 1 - \Phi(0) = \frac{1}{2}.$$

Another way to see the answer is the following: if X, Y, Z are mean zero then they have the same distribution as $-X, -Y, -Z$, respectively. Then $X + 2Y - 3Z$ has the same distribution as $-X - 2Y + 3Z$ which means that

$$P(X + 2Y - 3Z > 0) = P(-X - 2Y + 3Z > 0) = P(X + 2Y - 3Z < 0).$$

Since $X + 2Y - 3Z$ is continuous, we have $P(X + 2Y - 3Z = 0)$ and hence

$$P(X + 2Y - 3Z > 0) = P(X + 2Y - 3Z < 0) \quad \rightsquigarrow \quad P(X + 2Y - 3Z > 0) = 1/2.$$

7.28 Because X_1, X_2, X_3 are jointly continuous, the probability that any two of them are equal is 0. This means that $P(X_1, X_2, X_3$ are all different$) = 1$. By the exchangeability of X_1, X_2, X_3 we have

$$\begin{aligned}P(X_1 < X_2 < X_3) &= P(X_2 < X_1 < X_3) = P(X_1 < X_3 < X_2)\\ &= P(X_3 < X_2 < X_1) = P(X_2 < X_3 < X_1) = P(X_3 < X_1 < X_2),\end{aligned}$$

where we listed all six possible orderings of X_1, X_2, X_3. By additivity the sum of the six probabilities is $P(X_1, X_2, X_3$ are all different$)$. From this we get that $P(X_1 < X_2 < X_3) = \frac{1}{6}$.

7.30 We can label the cards of the deck with the integers from 1 to 52. Because we are sampling without replacement, the five random variables corresponding to the five picks are exchangeable.

(a) By exchangeability we can change the 2nd and 4th cards to 1st and 2nd in the event without changing the probability:

$$P(\text{2nd card is A, 4th card is K}) = P(\text{1st card is A, 2nd card is K}) = \frac{4 \cdot 4}{52 \cdot 51} = \frac{4}{663},$$

where the final probability comes from counting the favorable outcomes for the first two picks.
(b) Again, by exchangeability and counting the favorable outcomes within the first two picks:

$$P(\text{1st card is } \spadesuit, \text{ 5th card is } \spadesuit) = P(\text{1st card is } \spadesuit, \text{ 2nd card is } \spadesuit) = \frac{\binom{13}{2}}{\binom{52}{2}} = \frac{1}{17}.$$

(c) We use exchangeability again.

$$\begin{aligned}P(\text{2nd card is K}|\text{last two cards are aces}) &= \frac{P(\text{2nd card is K, last two cards are aces})}{P(\text{last two cards are aces})}\\ &= \frac{P(\text{3rd card is K, first two cards are aces})}{P(\text{first two cards are aces})}\\ &= P(\text{3rd card is K}|\text{first two cards are aces})\\ &= \frac{4}{50} = \frac{2}{25}.\end{aligned}$$

The final probability comes either from counting favorable outcomes for the first three picks, or by noting that if we choose two aces for the first two picks then we always have 50 cards left with 4 of them being kings.